**Laxmi Charitable Trust's**
**Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce**Department of Information Technology (B.Sc.I.T Semester V)

Applied Business Analytics

# Task

| Roll No.: T050 | Name: Aashu Yadav |
|---|---|
| Class: TYIT | Batch: 1 |
| Date of Assignment: 24/07/26 | Date/Time of Submission: 24/07/26 |

Task (on house_price_regression_dataset.csv) (Don't include in print document)

Q.1 Find the correlation between Year_Built and House_Price.

**code:**

```
import pandas as pd
df = pd.read_csv("house_price_regression_dataset - house_price_regression_dataset.csv")
correlation = df["Year_Built"].corr(df["House_Price"])
print("Correlation between Year_Built and House_Price:", correlation)
```

**Output:**

```
Correlation between Year_Built and House_Price: 0.051967361931064486
```

Q.2 Create a correlation matrix for all numerical columns.

**Code:**

```
import pandas as pd
df = pd.read_csv("house_price_regression_dataset - house_price_regression_dataset.csv")
correlation_matrix = df.corr(numeric_only=True)
print("Correlation Matrix:")
print(correlation_matrix)
```

**Output:**

```
Correlation Matrix:
                     Square_Footage  Num_Bedrooms  Num_Bathrooms  Year_Built  \
Square_Footage             1.000000     -0.043564      -0.031584   -0.022392
Num_Bedrooms              -0.043564      1.000000       0.022848   -0.015820
Num_Bathrooms             -0.031584      0.022848       1.000000   -0.021063
Year_Built                -0.022392     -0.015820      -0.021063    1.000000
Lot_Size                   0.089479     -0.009355       0.034923   -0.061050
Garage_Size                0.030593      0.113761       0.024846   -0.025485
Neighborhood_Quality      -0.008357     -0.049024       0.017585   -0.009549
House_Price                0.991261      0.014633      -0.001862    0.051967

                     Lot_Size  Garage_Size  Neighborhood_Quality  House_Price
Square_Footage       0.089479     0.030593             -0.008357     0.991261
Num_Bedrooms        -0.009355     0.113761             -0.049024     0.014633
Num_Bathrooms        0.034923     0.024846              0.017585    -0.001862
Year_Built          -0.061050    -0.025485             -0.009549     0.051967
Lot_Size             1.000000     0.002436              0.037630     0.160412
Garage_Size          0.002436     1.000000             -0.011287     0.052133
Neighborhood_Quality 0.037630    -0.011287              1.000000    -0.007770
House_Price          0.160412     0.052133             -0.007770     1.000000
```

Q.3 Perform linear regression using:
X = Year_Built
Y = House_Price
Display the coefficient and intercept.
**Code:**

```
import pandas as pd
from sklearn.linear_model import LinearRegressio
df = pd.read_csv("house_price_regression_dataset - house_price_regression_dataset.csv")
X = df[["Year_Built"]]
Y = df["House_Price"]
model = LinearRegression()
model.fit(X, Y)
print("Coefficient:", model.coef_[0])
print("Intercept:", model.intercept_)
```

**Output:**

```
Coefficient: 638.6524884727302
Intercept: -649854.0823295021
```

Q.4 Calculate the R2 score for the regression model:
Year_Built - House_Price
**Code:**

```
import pandas as pd
from sklearn.linear_model import LinearRegression
from sklearn.metrics import r2_score
df = pd.read_csv("house_price_regression_dataset - house_price_regression_dataset.csv")
X = df[["Year_Built"]]
Y = df["House_Price"]
model = LinearRegression()
model.fit(X, Y)
Y_pred = model.predict(X)
r2 = r2_score(Y, Y_pred)
print("R2 Score:", r2)
```

**Output:**

```
R2 Score: 0.0027006067060743044
```

Q.5 Calculate the R2 score for:
Square_Footage - House_Price
**Code:**

```
import pandas as pd
from sklearn.linear_model import LinearRegression
from sklearn.metrics import r2_score
df = pd.read_csv("house_price_regression_dataset - house_price_regression_dataset.csv")
X = df[["Square_Footage"]]
Y = df["House_Price"]
model = LinearRegression()
model.fit(X, Y)
Y_pred = model.predict(X)
r2 = r2_score(Y, Y_pred)
print("R2 Score:", r2)
```

**Output:**

```
R2 Score: 0.9825992634147099
```

Q.6 Calculate the R2 score for:
Neighborhood_Quality - House_Price

**Code:**

```
import pandas as pd
from sklearn.linear_model import LinearRegression
from sklearn.metrics import r2_score
df = pd.read_csv("house_price_regression_dataset - house_price_regression_dataset.csv")
X = df[["Neighborhood_Quality"]]
Y = df["House_Price"]
model = LinearRegression()
model.fit(X, Y)
Y_pred = model.predict(X)
r2 = r2_score(Y, Y_pred)
print("R2 Score:", r2)
```

**Output:**

```
R2 Score: 6.037338916142776e-05
```

Q.7 Compare the R2 scores of all three regression models.

| Regression Model | $R^2$ Score | Percentage Explained |
|---|---|---|
| Year_Built → House_Price | 0.0027 | 0.27% |
| Square_Footage → House_Price | 0.9826 | 98.26% |
| Neighborhood_Quality → House_Price | 0.0000604 | 0.01% |